10/28/24

CHAPTER 5 : Eigenvalues & Eigenvectors

§ 5.1 Systems of differential equations

(motivation for studying eigenvalues)

Consider $x_1(t)$, $x_2(t)$: "nice" functions of "time" $t \in \mathbb{R}$, $(t \geq 0)$. Suppose

$$(*) \quad \begin{cases} \dot{x}_1(t) = 2x_1(t) + 3x_2(t) \\ \dot{x}_2(t) = x_1(t) + 4x_2(t) \end{cases}, \quad t \geq 0$$

Task: Solve this system of d.e's for $x_1(t)$, $x_2(t)$

Linear Algebraic approach:

Define $\dot{x} = \dot{x}(t) = \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix}$

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

$$x = x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

Rewrite $(*)$

$$\boxed{\dot{x} = Ax, \quad t \geq 0} \quad (LA)$$

linear autonomous

<u>Solution</u> uses the notion of "matrix exponential":

$$e^{tA} := I + tA + \frac{(tA)^2}{2!} + \cdots = \sum_{k=0}^{\infty} \frac{t^k A^k}{k!}$$

(infinite series) – $t \geq 0$

This infinite series converges for all $A \in M_n(\mathbb{R})$, $t \geq 0$:

$$\| e^{tA} \| = \left\| \sum_0^{\infty} \frac{t^k A^k}{k!} \right\| \leq \sum_0^{\infty} \frac{t^k \|A^k\|}{k!}$$

$$\underset{\text{submultiplicativity}}{\leq} \sum_0^{\infty} \frac{t^k \|A\|^k}{k!} = \underset{\text{scalar function}}{e^{t\|A\|}} < \infty$$

$$\frac{d}{dt}\left(e^{tA}\right) = \frac{d}{dt}\left(\sum_{k=0}^{\infty} \frac{t^k A^k}{k!}\right)$$

$$= \sum_{k=1}^{\infty} \frac{k t^{k-1} A^k}{k!}$$

$$= A \sum_{k=1}^{\infty} \frac{\cancel{k} t^{k-1} A^{k-1}}{\cancel{k!}\,(k-1)!}$$

$$= A \sum_{k=0}^{\infty} \frac{t^k A^k}{k!} = A e^{tA}$$

Thus $x(t) = e^{tA} x(0)$, $t \geq 0$

is a solution to (LA) with ~~the~~ any initial condition $x(0)$ at $t=0$:

$$\dot{x}(t) = \frac{d}{dt}\left(e^{tA} x(0)\right)$$

$$= A e^{tA} x(0) = A x(t)$$

$$\& \quad x(0) = e^{0A} x(0) = x(0) \checkmark$$

<u>Issue</u> How do we compute e^{tA}?
Truncation of the infinite series does <u>not</u> work for $t \gg 0$.

Eigenvalues enter the picture
(nilpotency of A, diagonalizability of A
: easy cases.)

[Famous paper in SIAM Review:
"50 dubious ways of finding the matrix exponential"
"updated".]

Matlab: $\boxed{\text{expm}(A)}$ (logm(A))

Easy cases: Nilpotent : $A^k = 0$ for some k
inf. series $\longrightarrow$ finite

Diagonalizable A:

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

$$\det(tI - A) = (t-1)(t-5)$$

Eigenvalues: $\lambda_1 = 1$, $\lambda_2 = 5$

Corr. eigenvectors: $v_1 = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

A is diagonalizable:

$$S = [v_1 \mid v_2] = \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}$$

$$S^{-1} A S = D$$

<u>Exploit</u>: $y := y(t) = S^{-1} x(t)$ (change of variables)

Then

$$\dot{y} = (S^{-1} x)^{\cdot} = S^{-1} \dot{x} = S^{-1} A x$$

$$= S^{-1} A \underline{S S^{-1}} x$$

$$= (S^{-1} A S) y$$

$$= D y$$

That is,

$$\begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$\begin{cases} \dot{y}_1 = y_1 \\ \dot{y}_2 = 5y_2 \end{cases}$$

So $\quad y_1 = c_1 e^{t}, \quad c_1 \in \mathbb{R}$

$\quad y_2 = c_2 e^{5t}, \quad c_2 \in \mathbb{R}$

Thus

$x(t)$ = solution to original problem

$(y = S^{-1}x)$

$$= Sy(t)$$

$$= \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} c_1 e^{t} \\ c_2 e^{5t} \end{bmatrix}$$

i.e.

$$= \begin{bmatrix} 3c_1 e^{t} + c_2 e^{5t} \\ -c_1 e^{t} + c_2 e^{5t} \end{bmatrix}$$

$$\boxed{\begin{aligned} x_1(t) &= 3c_1 e^{t} + c_2 e^{5t} \\ x_2(t) &= -c_1 e^{t} + c_2 e^{5t} \end{aligned}}$$

(if we know $x_1(0)$, $x_2(0)$: find c_1, c_2)

This exemplifies the importance of "eigenstuff".

<u>Comment</u> If A is <u>not</u> diagonalizable, "use" Jordan Canonical Form!

$(S^{-1}AS = J)$ A is $\boxed{\text{similar}}$ to a Jordan matrix J

& $J = N + D$, N = nilpotent, D = diagonal & $DN = ND$

always can

Then

$$e^{A} = e^{SJS^{-1}} \underset{\text{carry}}{=} Se^{J}S^{-1}$$

$$= Se^{N+D}S^{-1}$$

$$= Se^{N}\cdot e^{D}S^{-1}$$

computable

computable

<u>fact!</u>

$e^{X+Y} = e^{X}e^{Y}$ iff $XY = YX$

However, computing/knowing J is challenging.

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§ 5.2 Basic Facts about eigenvalues and eigenvectors

$A \in M_n(\mathbb{C})$

$Av = \lambda v$, $v \in \mathbb{C}^n \setminus \{0\}$, $\lambda \in \mathbb{C}$

v → eigenvector

λ → eigenvalue

(λ, v) eigenpair

spectrum of A

$$\sigma(A) = \{\lambda \in \mathbb{C} : \lambda \text{ eigenvalue}\}$$

Questions: 1) existence of eigenpairs?

2) how do we find eigenpairs?

Viewpoint: $\underset{v \neq 0}{Av = \lambda v} \iff \underset{v \neq 0}{(A - \lambda I)v = 0}$

$\iff A - \lambda I$ is not invertible

$\iff \det(A - \lambda I) = 0$

$\iff \boxed{\det(\lambda I - A) = 0}$

characteristic polynomial

characteristic equation

- Characteristic equation is an equation in λ of polynomial type of degree n (monic)
  $\downarrow$
  coeff. of $\lambda^n = 1$

- Solving the charact. equation allows to find the eigenvalues:

  n roots $\longrightarrow$ n eigenvalues

- Since $\lambda \in \sigma(A) \iff Av = \lambda v$ for some $v \neq 0$, this proves existence of eigenpairs.

- The multiplicity of $\lambda \in \sigma(A)$ as a root of char. polynomial is known as the

  <u>algebraic multiplicity of λ</u> $= am(\lambda)$

Since the nullspace $N(A - \lambda I)$ is nontrivial subspace if $\lambda \in \sigma(A)$, it has a basis (of eigenvectors & a dimension); known as the <u>geometric multiplicity of λ</u>:

$gm(\lambda)$

- To find eigenpairs
  1) Find eigenvalues λ (solve $\det(\lambda I - A) = 0$)
  2) $\forall \lambda \in \sigma(A)$, solve
  $$(A - \lambda I)\, v = 0,$$
  a basis of solution space is a basis for $N(A - \lambda I)$ which we call the <u>eigenspace</u> corr. to $\lambda \in \sigma(A)$.

- Requires (non-trivial) proof
  $$\boxed{am(\lambda) \geq gm(\lambda)}$$

- If $A \in M_n(\mathbb{R})$, then $\det(\lambda I - A)$ is a <u>real</u> polynomial. So its roots come in complex conjugate pairs:
  $$\lambda \in \sigma(A) \setminus \{\mathbb{R}\} \iff \bar{\lambda} \in \sigma(A) \setminus \{\mathbb{R}\}$$
  $$\lambda = a + bi \longleftrightarrow \bar{\lambda} = a - bi$$
  $$b \neq 0 \qquad\qquad b \neq 0$$

eg. $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ $\quad \det(\lambda I - A) = \lambda^2 + 1 = 0$

$\lambda = \pm i$

- Issue: $\det(tI - A) = 0$ is ok to solve explicitly ("exactly") if $n \leq 4$. But $n \geq 5$ it cannot be done (Algebra)

So now what?

Answer:
- Approximate eigenvalues
- Numerical methods
- Matrix methods are all iterative in nature involving the powers of A^k, $k = 1, 2, \ldots$

<u>Theorem</u> Let $A = \begin{bmatrix} \boxed{A_{11}} & & * \\ & \boxed{A_{22}} & \\ 0 & & \ddots \; \boxed{A_{kk}} \end{bmatrix} \in M_n(\mathbb{C})$

where A_{jj} are square of size $k_j \times k_j$ with $\sum_{j=1}^{k} k_j = n$. (Block-triangular matrix)

Then $\sigma(A) = \sigma(A_{11}) \cup \sigma(A_{22}) \cup \ldots \cup \sigma(A_{kk})$

(Note: "sets of eigenvalues" are multisets in our discussions:

$\sigma(I_3) = \{1, 1, 1\}$)

<u>Proof</u> First $\det(\lambda I - A) \ldots$

<u>Theorem</u> IF $v_1, v_2, \ldots, v_k \in \mathbb{C}^n$ are the eigenvectors of $\lambda_1, \lambda_2, \ldots, \lambda_k \in \sigma(A)$ for some $A \in M_n(\mathbb{C})$, and if

$$\lambda_1, \lambda_2, \ldots, \lambda_k \in \sigma(A)$$

are <u>distinct</u>, then $\{v_1, v_2, \ldots, v_k\}$ is a linearly independent set.

<u>Proof</u> Work with a lin. comb. of $v_1, \ldots, v_k$ and hit it with A ... ∎

<u>Consequences</u>

- if all eigenvalues of A are distinct, the A has "complete set" of lin. eigenvectors and so it <u>diagonalizable</u>:

$$A v_j = \lambda_j v_j, \quad \lambda_j \neq \lambda_i \quad i \neq j = \{1, \ldots, n\}$$

$$S = [v_1 | v_2 | \ldots | v_n] \leftarrow \text{invertible}$$

$$S^{-1} A S = D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} = \text{diagonal}$$

- <u>Not</u> all matrices are diagonalizable

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \sigma(A) = \{1, 1\}$$

$$\dim N(A - I) = 1 = gm(1) < am(1) = 2$$